AI STUDY BUDDY

Idea submission for AI Unlocked

Institute: Indian Institute Of Technology Kharagpur

Challenge Area: AI STUDY BUDDY

Your Team Name: LazyBuild

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Project Title - MentorMate

Project subtitle: Multimodal AI for Handwritten Note Digitization & Multilingual Learning.

The Team

Our team is a group of third-year undergraduate students from IIT Kharagpur who came together through a shared passion for leveraging artificial intelligence to improve learning.
Each member brings a unique set of skills and expertise to the project:
Harsh Dalmia brings expertise in DevOps and deployment, ensuring scalable and reliable system infrastructure.
Krish Sahu focuses on artificial intelligence and model development, building the core intelligence behind note analysis and concept understanding.
Harsh Singh works on artificial intelligence and system integration to enable seamless interaction among multimodal AI components.
Ayush Kumar specializes in product strategy and user experience, focusing on designing student-centric learning workflows.

The Concept

Every exam season brings the same frustration, messy handwritten notes, confusing diagrams, and concepts that once felt clear but now seem impossible to understand. Students spend hours rewriting notes or watching random videos, yet still feel lost, especially when they study in English but think in their native language. Many fall behind not because they aren't capable, but because they don't have personalized guidance.

MentorMate solves this by turning handwritten notes into a personal AI tutor. Students upload a photo, and MentorMate uses Azure Vision to read messy handwriting, GPT-4o with Vision to simplify concepts, and Azure Speech to explain them aloud in English and their preferred Indian language. It also creates quizzes and tracks weak topics, transforming confusing notes into clear understanding and making studying engaging, personalized, and accessible for millions of students.

Target Audience or Market

Our primary target audience is undergraduate engineering students who rely on handwritten notes, diagrams, and classroom explanations for exam preparation. Many struggle to revise messy notes, understand complex concepts, and access personalized academic support.
MentorMate is designed for students in technical institutes where subjects involve dense problem solving and visual content. It is accessible via web and mobile for flexible learning.
The initial focus is India, where linguistic diversity creates learning barriers. With over 3,500 engineering institutions and 15 million students, the market is significant. The solution can further scale globally to support students facing similar challenges.

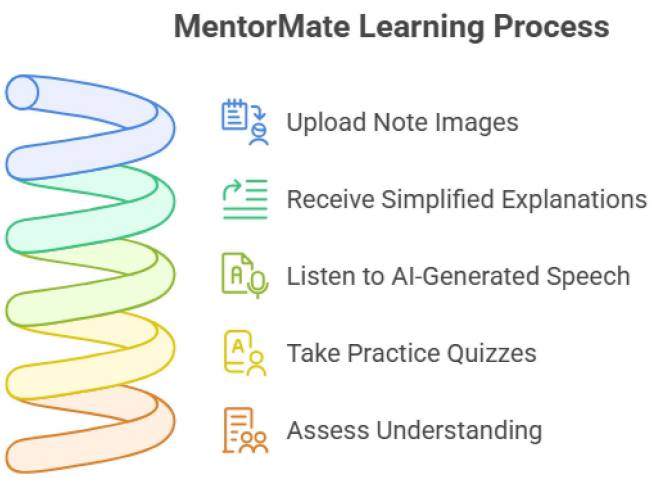
Personas

Persona 1: Exam-Focused Engineering Student

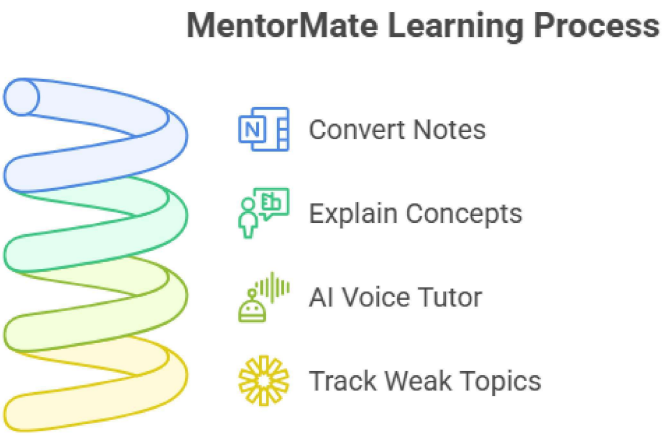
Name: Rohan Sharma
Age: 20
Education: Third-Year Engineering Student
Location: India

Relies heavily on handwritten notes for exam preparation. Often struggles to revise messy diagrams, formulas, and rushed classroom explanations.

Needs structured summaries, simplified concept breakdowns, and quick revision support generated directly from his notes through AI assistance.



Persona 2: Multilingual Concept Learner



Name: Ananya Verma
Age: 19
Education: Second-Year Engineering Student
Location: Tier 2 City, India

Understands concepts better in her native language but studies mostly from English material, making technical subjects harder to grasp.

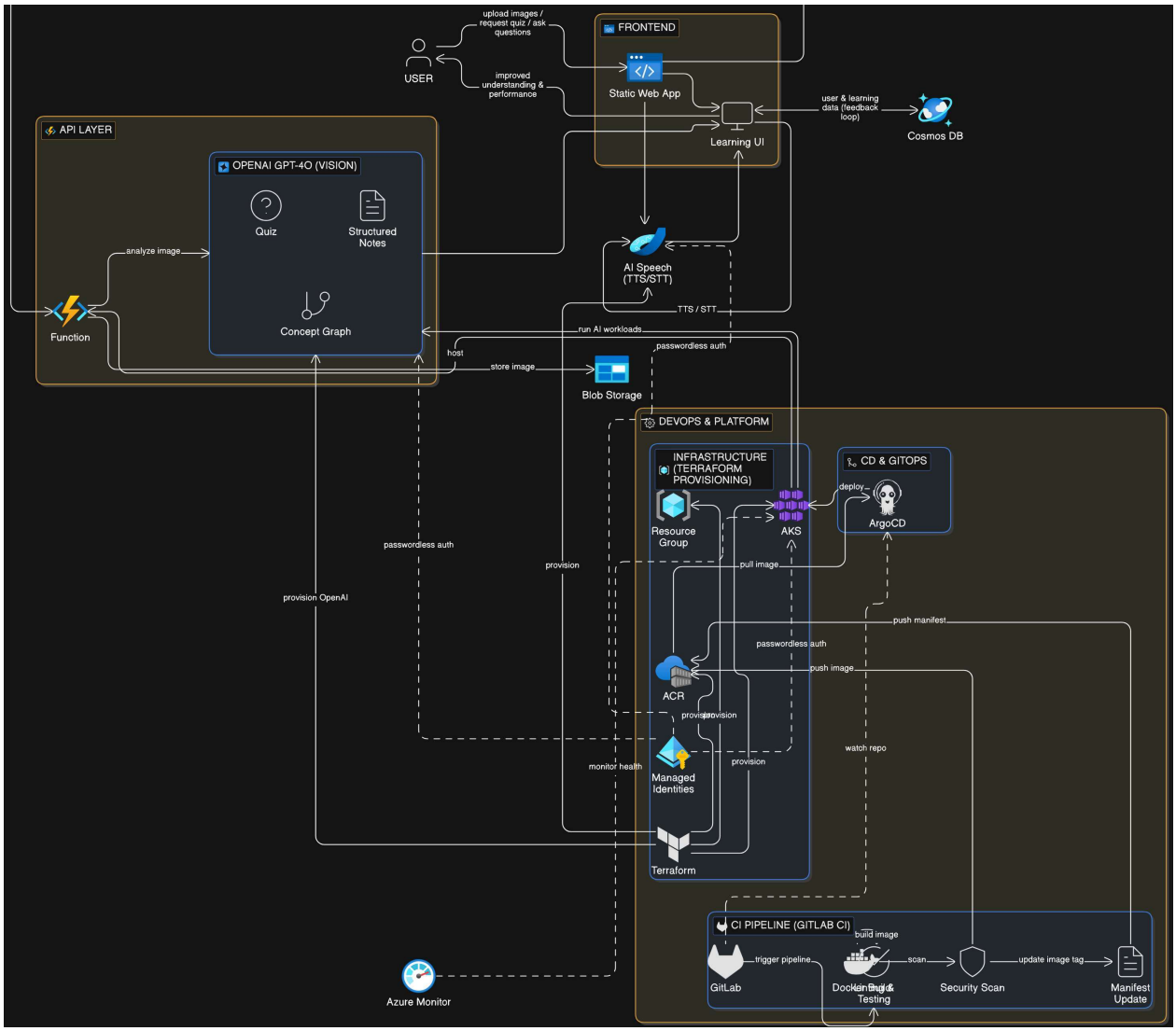
Needs multilingual explanations, audio learning support, and personalized quizzes to improve

How it Works

How We Built It: The Microsoft Advantage

Microsoft Azure provided the **multimodal backbone** necessary to bridge physical notes and digital learning in a single, high-performance ecosystem.

- **Azure OpenAI (GPT-4o with Vision):** Eliminated the need for separate OCR and NLP pipelines. A single API call processes handwritten diagrams, extracts logic, and generates bilingual quizzes.
- **Azure AI Speech:** Provided the linguistic depth needed for the Indian market, offering high-fidelity **Neural TTS** in regional languages for natural-sounding auditory learning.
- **Azure Kubernetes Service (AKS):** Offered a scalable, managed environment to host the AI orchestration logic, ensuring low-latency responses for real-time user interactions.
- **Azure Container Registry (ACR):** Secured our Dockerized microservices with integrated vulnerability scanning.



Core Technologies

AI & Data Processing

- **OpenAI GPT-4o (Vision):** The central engine for multimodal analysis, responsible for transforming image requests into structured notes, concept graphs, and quizzes.
- **AI Speech (TTS/STT):** Provides the audio interface layer for speech-to-text and text-to-speech interactions.
- **Azure Functions:** Acts as the serverless compute layer to host and execute API logic and AI workload triggers.

Frontend & Application Hosting

- **Azure Static Web Apps:** Provides the hosting environment for "Learning UI" application.
- **Cosmos DB:** A globally distributed database used for managing user data and maintaining the learning feedback loop.
- **Blob Storage:** Scalable object storage used for uploaded user images and hosted assets.

Platform & Infrastructure

- **Azure Kubernetes Service (AKS):** The primary orchestration platform for running and scaling the containerized AI workloads.
- **Terraform:** The Infrastructure as Code (IaC) tool used for the automated provisioning of Azure resources and managed identities.
- **Azure Container Registry (ACR):** A private registry used to store and manage the Docker images required for deployment.

CI/CD & GitOps

- **GitLab CI:** The foundation of the continuous integration pipeline, handling automated builds, security scans, and testing.
- **ArgoCD:** A GitOps-based tool for continuous delivery, ensuring the AKS cluster remains synchronized with the latest manifests.

The Business Plan – MentorMate

MentorMate will launch as a freemium AI study companion designed for students struggling with messy handwritten notes and complex concepts. The free version will offer limited scans, explanations, and quizzes, while a premium plan priced at ₹149 per month or ₹999 per year will unlock unlimited scans, multilingual audio tutoring, weak topic tracking, and exam mode mock tests, keeping it affordable compared to private tutors. MentorMate will also partner with schools and coaching centers to provide teacher dashboards, test generation, and progress tracking. Through campus launches, institutional collaborations, and regional language support, our goal is to create meaningful learning impact by turning confusion into confidence for students.